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UI-AT3

Descriptive

Test

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1. A smart home automation system uses AI to automatically controls lighting, temperatures and security based on user behaviours and environmental condition.

Types of Intelligent Agents

I Simple reflex Agent

- * Works only on the current input.
- * Uses predefined if-then rules
- * Does not remember previous events.

Example:

If motion is detected → Turns on the light
If no motion is detected → Turns off the light

Advantages:

- * Fast response & easy to implement
- * Suitable for simple tasks

Disadvantages:

- * Cannot learn user habits.

II. Model - Based Reflex agent.

- * Maintains an internal model of the environment.
- * Uses both current and previous info.

Advantages

- * Makes better decision using memory.

Disadvantages.

- * Requires additional storage processing.

III. Goal - Based Agent

- * Makes decision based on achieving specific goals.

- * Plans action to reach the desired outcome.

Advantages:

- * Flexible and can plan multiple actions

Disadvantages

- * Planning takes more computation time.

IV Utility based Agent

* Selects the action that gives the highest overall benefit.

Advantages.

- * Makes the best overall decision
- * Handles conflicting objectives

Disadvantages.

- * More complex to design.

Agent type	Memory	Goal	Learning	Best use.
Simple Reflex Model	No	No	No	Basic lighting
- Based	Yes	No	limited	Environment monitoring
Goal - Based	Yes	Yes	No	Planning Tasks.
Utility based	Yes.	Yes.	Yes	Comfort, energy and security

Best choice: A hybrid Agent combining model, goal and utility based agent.

Reason:

- * learns user activity.
- * Saves Electricity
- * Improves comfort.

2. A robot placed in an unknown maze must search for a path to the exit without knowing the maze layout.

Since no prior information is available, uniformed search algorithm (UCS) and Iterative Deepening Search (IDS) are

Suitable.

Uniform Cost Search (UCS)

working:

- * Expands the node with the lowest

* Uses a priority queue.

Advantages.

* Find the path

+ Suitable for weighted maps.

Disadvantages.

+ High memory usage

* Can be slow for large maps.

Complexity.

$$\text{Time: } O^*(b^l(1 + c^*/\epsilon))$$

$$\text{Space: } O(b^l(1 + c^*/\epsilon))$$

where

b = branching factor

c^* = optimal path cost.

ϵ = minimum step cost.

Iterative Deepening Search (IDS)

working.

- * Performs depth first search repeatedly with increasing limits.

Advantages.

- * Low memory usage and suitable when goal depth is unknown.

Disadvantages.

- * Repeats node expansion.
- * Slower than BFS in some cases.

Complexity

* Time : $O(b^d)$

* Space : $O(b \times d)$

where

b = branching factor.

d = depth of solution

Relation to intelligent agents

Simple reflex: Not suitable.

Model Based Agent: Suitable for maze exploration.

Goal Based agent: Selects path that move closer to the goal.

Utility based Agent: Consider multiple factors while moving.

Effect of agent design:

* Agents with memory avoid repeated exploration.

* Goal Based Agents focus on reaching the exit.

* Utility based agents optimize travel cost and safety.

Best combination: Goal based + Mode

Base using UCS and IDS

Justification

- * Model based agent remembers explored paths.
- * Goal Based Agent keeps the robot focused on finding the exit.
- * UCS guarantees the least cost path.
- * IDS provides a memory efficient.